

CURRICULUM VITAE

YVES KIMBELE HERI

Phone: +1-517-329-5115
Email: heriyves@umich.edu
Website: <https://www.yves-heri.com/>
GitHub: <https://github.com/yvesheri4>

RESEARCH INTERESTS

Ultrafast Electron Beam Dynamics and Beam–Plasma Acceleration

Short-pulse electron beam transport in vacuum and plasma, emphasizing space-charge effects (pulse broadening, energy spread, phase-space distortion) and beam–plasma interactions in high-energy-density regimes.

Sheath Physics and Plasma–Surface Interactions

Modeling emission-driven plasma sheaths using fluid and kinetic approaches, including inverse sheath formation and particle flux redistribution at material interfaces relevant to fusion devices.

Computational Plasma Modeling

Development of fluid and kinetic plasma solvers, numerical methods, and data-driven approaches, including machine learning for reduced-order modeling and predictive simulations of low-temperature and high-energy-density plasmas.

EDUCATION

Ph.D. Student, Nuclear Engineering and Radiological Sciences University of Michigan (Plasma, Beams, and Interface Science Group, Prof. Peng Zhang)	Jan 2025 – Present Ann Arbor, MI, USA
Master of Science in Electrical and Computer Engineering Michigan State University	Aug 2022 – Dec 2024 East Lansing, MI, USA
Master of Science in Mathematical Sciences African Institute for Mathematical Sciences	Aug 2021 – Jun 2022 Kigali, Rwanda
Master of Science in Electrical Engineering University of Lubumbashi	Oct 2015 – Jul 2017 Lubumbashi, Congo DR
Bachelor of Science in Electrical Engineering University of Lubumbashi	Oct 2012 – Jul 2015 Lubumbashi, Congo DR

ADDITIONAL EDUCATION

Graduate Certificate in Accelerator Science and Engineering Michigan State University, Department of Physics and Astronomy	Aug 2022 – Dec 2024 East Lansing, MI, USA
Graduate Certificate in Computational Modeling Michigan State University, Computational Mathematics, Science and Engineering	Aug 2022 – Dec 2024 East Lansing, MI, USA
Graduate Certificate in Semiconductor Devices and Manufacturing Michigan State University, Electrical and Computer Engineering	Aug 2022 – Dec 2024 East Lansing, MI, USA
High Energy Density Science Summer School University of California, San Diego	Jul 2023 La Jolla, CA, USA
U.S. Low Temperature Plasma Summer School University of Michigan	Jun 2023 Ann Arbor, MI, USA

PROJECTS AND RESEARCH

Computational Plasma Physics <i>PhD</i> Short-pulse beam dynamics; space-charge effects; PIC (WarpX)	Jan 2025 – Present University of Michigan
Electricity Theft Detection <i>M.Sc.</i> Machine learning for anomaly detection in power data	Jan 2022 – Jun 2022 AIMS Rwanda
Optimized PV System Design <i>M.Sc.</i> Photovoltaic system design and performance analysis	Oct 2017 – Apr 2018 University of Lubumbashi
PID Tuning Methods <i>B.Sc.</i> Comparison of classical PID tuning techniques	Apr 2015 – Dec 2015 University of Lubumbashi

PUBLICATIONS AND CONFERENCES

Journal Articles

- Y. Heri** and P. Zhang, "Space Charge Effects on Short-Pulse Beam Dynamics in a Drift Space," *IEEE Transactions on Electron Devices*, 2026. doi: 10.1109/TED.2026.3670804
- Y. Heri** and P. Zhang, "Space Charge Effects on Short-Pulse Electron Beam Dynamics in a Classical Vacuum Diode," *IEEE Transactions on Electron Devices*, 2025. doi: 10.1109/TED.2025.3552061
- A. Iqbal, **Y. Heri**, B. Wang, L. Jin, M. A. Faisal, and P. Zhang, "PASCHEN-1D: A One-Dimensional Fluid Plasma Solver with Multi-Mechanism Surface Emission and Flexible External Circuit Coupling," *Computer Physics Communications*, under review (2026).

Conference Papers

- Y. Heri** and P. Zhang, "Space Charge Effects on Short-Pulse Electron Beam Dynamics," *38th International Vacuum Nanoelectronics Conference (IVNC), Reykjavik, Iceland*, 2025. doi: 10.1109/IVNC65669.2025.11120901
- Y. Heri** and P. Zhang, "Space Charge Limited Current Scaling for Short-Pulse Beam in a Vacuum Diode with Different Pulse Shapes," *IVEC + IVESC, Monterey, CA, USA*, 2024. doi: 10.1109/IVECIVESC60838.2024.10694914
- Y. Heri** and P. Zhang, "Space charge effects on the short pulse beam profile," *IEEE International Conference on Plasma Science (ICOPS), Santa Fe, NM, USA*, 2023. doi: 10.1109/ICOPS45740.2023.10480963
- Y. Heri** and B. W. B. Bonaventure, "Price Dynamics for Net Electric Energy Metering on a Distribution Network: Ising Model Approach," *IEEE PES/IAS PowerAfrica, Nairobi, Kenya*, 2021, pp. 1–5. doi: 10.1109/PowerAfrica52236.2021.9543305

WORK EXPERIENCE

Graduate Research Assistant Short-pulse beam dynamics; space-charge effects; PIC (WarpX)	Jan 2025 – Present University of Michigan, Ann Arbor, MI, USA
Graduate Research Assistant Charged particle modeling; ML for plasma surrogate models	Aug 2022 – Dec 2024 Michigan State University, East Lansing, MI, USA
Assistant Lecturer Teaching and supervision in electrical engineering	May 2018 – Present University of Lubumbashi, Lubumbashi, Congo DR
Intern Engineer Industrial electrical system maintenance	May 2017 – Oct 2017 CHEMAF Usoke, Lubumbashi, Congo DR

HONORS AND AWARDS

Inductee, Alpha Nu Sigma National Honor Society (UM Student Section) Recognized for academic excellence, integrity, and potential in Nuclear Engineering.	Feb 2026
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Best Presentation Award, 14th MIPSE Graduate Symposium Awarded for best poster presentation at a competitive graduate symposium in Ann Arbor, MI	Nov 2023
AIMS Rwanda Excellence Award Awarded to the top-performing student at the African Institute for Mathematical Sciences	Jun 2022
MasterCard Foundation Scholar Competitive scholarship recognizing academic excellence, leadership, and commitment to socio-economic development	Feb 2022
AIMS Rwanda Scholarship Merit-based scholarship to pursue an M.S. in Mathematical Sciences	Aug 2021
MMG Scholarship for Academic Excellence Merit-based scholarship awarded to high-achieving engineering students at the University of Lubumbashi	2016
Best Laureate in Electronics, Katanga State Exam Ranked first in the provincial state examination in Electronics	2012

TEACHING EXPERIENCE

Programmable Logic Systems (ESI) University of Lubumbashi	2020 – 2021 Lubumbashi, Congo DR
Power Distribution Networks (ESI) University of Lubumbashi	2019 – 2020 Lubumbashi, Congo DR
Physics – Electromagnetism University Katumba-Mwanke	2018 – 2019 Kasenga, Congo DR

COMMUNITY INVOLVEMENT

Mount Hope Tutoring Program Assisting public school students with weekly assignments and academic support	Nov 2022 – Dec 2025 Lansing, MI, USA
Head, Scientific Department, Cultural Center of ESI Organized lectures and events with guest researchers and professionals	2015 – 2017 Lubumbashi, Congo DR
Soccer Team Assistant Manager Coordinated and supervised youth participation in local tournaments	2015 Lubumbashi, Congo DR
Volunteer Teacher, Programming and Robotics Taught programming and robotics to students from elementary to university level	2020 – Present Lubumbashi, Congo DR

SKILLS

Languages: English, French (native), Swahili

Programming: Python (NumPy, Pandas, PyTorch, TensorFlow, Keras, scikit-learn), C, C++, MATLAB

Simulation Tools: WarpX, OSIRIS, XOOPIC, CST Studio Suite

Document Preparation: L^AT_EX, Markdown, Microsoft Office Suite